



Cambridge IGCSE™

COMPUTER SCIENCE

0478/21

Paper 2

October/November 2020

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2020 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **6** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Question	Answer	Marks
Section A		
1(a)	For one array One mark for naming of an array with a suitable data type and one mark for its use Array yearsWithoutClaim Data type integer Use to store the % discount Many correct answers, this is an example only.	3
1(b)	For each validation check One mark for identification of a check and one mark for the reason max 4 Check presence check Reason need to enter age to calculate price Check type check Reason need to check age is a positive integer Many correct answers, these are examples only.	4
1(c)	Any six from: MP1 Input with prompt if new customer MP2 Check age greater than or equal 26 MP3 Check age less than or equal 70 MP4 Check number of years without a claim greater than or equal to 2 ... MP5 ... discount 10% MP6 Otherwise discount 0% MP7 Calculate new price and value of extra discount MP8 Output new price and value of extra discount, with suitable message Sample answer PRINT "Are you a new customer Y/N?" INPUT new discount ← 0 IF new = "Y" DO IF age ≥ 26 AND age ≤ 70 THEN IF yearsWithoutClaim ≥ 2 THEN discount ← price * 0.1 price ← price * 0.9 ENDIF ENDIF PRINT "New price", price PRINT "Extra discount for new customer", discount ENDIF	6

Question	Answer	Marks
1(d)(i)	Explanation Any four from: MP1 Provide a prompt and input to check if additional driver required MP2 Check response with IF/REPEAT/WHILE/CASE statement for yes MP3 ... input age of additional driver MP4 ... Use of IF / REPEAT etc. to validate age MP5 ... use the age input to find / look up appropriate price increase MP6 ... description of how the price to pay now is calculated MP7 ... description of how the price to pay now is displayed including the use of a suitable message For no MP8 ... no output required or suitable message output Programming statements can be used but must be explained.	4
1(d)(ii)	Description Any three from: MP1 Use a variable (quotes) to keep a running total of the number of quotes provided MP2 Ask, using prompt with input, customers if they are going to purchase the insurance MP3 Use another variable (policies) to keep a running total of the number of insurances bought / not bought MP4 Use these to calculate the percentage policies/quotes*100	3

Question	Answer	Marks
Section B		
2	One mark for description of variable One mark for description of constant One mark for inclusion of an example variable One mark for inclusion of an example constant Example answer: A value that can change during the execution of a program A named value that cannot change during the execution of a program Variable example - using a counter for example <code>counter ← counter + 1</code> Constant example – a static value that can be used for checking for example <code>< maxAge</code>	4

Question	Answer	Marks
3(a)	Line 1/2/3/4/8/9 Lines 5–11 Line 9 Line 8	4

Question	Answer	Marks
3(b)	One mark for error and correction Line 1 <code>TotalWeight ← 0</code> Line 10 move <code>OUTPUT "Number of bags in the load is ", BagCount</code> to end / after line 11 Line 11 <code>UNTIL TotalWeight > MaxWeight</code> OR <code>BagCount >= MaxBag</code> Line 12 <code>OUTPUT "Total weight of the load is ", TotalWeight</code>	4
3(c)	Any four from: After line 11 Divide <code>TotalWeight</code> by ... <code>BagCount</code> Assign a new variable <code>AverageWeight ← TotalWeight / BagCount</code> Output the result <code>OUTPUT AverageWeight</code> With a message <code>"Average weight of a bag of firewood is "</code>	4

Question	Answer	Marks																																								
4(a)	One mark for correct input columns (V, W, X, Y, Z) One mark for correct calculation column A One mark for correct calculation column B One mark for correct output column <table><tr><th>V</th><th>W</th><th>X</th><th>Y</th><th>Z</th><th>A</th><th>B</th><th>OUTPUT</th></tr><tr><td>5</td><td>4</td><td>6</td><td>2</td><td>1</td><td>56</td><td>1</td><td>Valid</td></tr><tr><td>9</td><td>3</td><td>2</td><td>1</td><td>6</td><td>40</td><td>7</td><td>Invalid</td></tr><tr><td>7</td><td>6</td><td>1</td><td>5</td><td>1</td><td>61</td><td>6</td><td>Invalid</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td></td><td></td><td></td></tr></table>	V	W	X	Y	Z	A	B	OUTPUT	5	4	6	2	1	56	1	Valid	9	3	2	1	6	40	7	Invalid	7	6	1	5	1	61	6	Invalid	0	0	0	0	0				4
V	W	X	Y	Z	A	B	OUTPUT																																			
5	4	6	2	1	56	1	Valid																																			
9	3	2	1	6	40	7	Invalid																																			
7	6	1	5	1	61	6	Invalid																																			
0	0	0	0	0																																						
4(b)	• (Use first four digits input to) calculate a check digit • Check if the check digit input is valid	2																																								

Question	Answer	Marks
5(a)	6 fields 9 records	2
5(b)	Either mark by row or by column but not both One each correct row or one mark for each correct column max two Koala3 Australia SkyKing Europe	2

Question	Answer				Marks																														
5(c)	<table><tr><td>Field:</td><td>Airline name</td><td>Number of countries</td><td>Head office</td><td></td></tr><tr><td>Table:</td><td>AIRLINE</td><td>AIRLINE</td><td>AIRLINE</td><td></td></tr><tr><td>Sort:</td><td></td><td></td><td></td><td></td></tr><tr><td>Show:</td><td>☒</td><td>☒</td><td>☐</td><td>☐</td></tr><tr><td>Criteria:</td><td></td><td>> 4</td><td>="Asia"</td><td></td></tr><tr><td>or:</td><td></td><td>> 4</td><td>="Africa"</td><td></td></tr></table> one mark for correct fields and correct table names, must have the three columns required, ignore another column provided the show row is not ticked and there are no criteria set for that column one mark for correct show row one mark for correct number of countries criteria one mark for correct head office criteria				Field:	Airline name	Number of countries	Head office		Table:	AIRLINE	AIRLINE	AIRLINE		Sort:					Show:	☒	☒	☐	☐	Criteria:		> 4	="Asia"		or:		> 4	="Africa"		4
Field:	Airline name	Number of countries	Head office																																
Table:	AIRLINE	AIRLINE	AIRLINE																																
Sort:																																			
Show:	☒	☒	☐	☐																															
Criteria:		> 4	="Asia"																																
or:		> 4	="Africa"																																